Learning long-range spatial dependencies with spectral convolutional state space models

Anonymous Author(s)

Abstract

Visual perception emerges from computations performed over feedforward and recurrent feedback connections between neurons. Yet today's state-of-the-art deep neural networks are almost entirely feedforward. Scaling has favored exclusively feedforward architectures that fit deep learning hardware, even at the cost of biological alignment. Here, we bridge the gap between biology and deep learning at scale with spectral convolutional state space models (SCSSMs), a scalable vision architecture that implements recurrent horizontal connections by lifting convolutional recurrence into the Fourier domain. A single SCSSM layer achieves human-level performance on visual reasoning challenges where transformers fail, and can be scaled up to large-scale vision benchmarks while matching transformer training speeds with fewer parameters and improved performance.

Recurrent neural networks are a central concept for explaining brains and behavior. In vision, local recurrent circuits through horizontal connections support contour integration, perceptual grouping, and figure-ground segmentation (Bosking et al., 1997; Field et al., 1993; Gilbert & Li, 2013; Gilbert & Wiesel, 1989; Mély et al., 2018). The horizontal gated recurrent unit (hGRU) (Linsley et al., 2018) demonstrated that these connections could be incorporated into a deep learning architecture and trained end-to-end to solve multiple perceptual challenges (Linsley et al., 2020). But like all nonlinear RNNs, the hGRU is inherently sequential, rendering training slow and poorly suited for the capabilities of GPUs and TPUs. In contrast, Transformers (Vaswani et al., 2017) are an exquisite fit for GPU and TPU capabilities and scaling, but struggle on tasks that demand structured long-range spatial integration (Tay et al., 2021) and have are often misaligned with human perception (Linsley et al., 2025).

State space models (SSMs) (Gu et al., 2022) are a potential bridge between the architectural preferences of today's deep learning hardware and the architectural features of brains: linear recurrences that admit efficient parallel training. But extending SSMs to vision data has required strict constraints that make it impossible to implement biologically inspired recurrent connections. The S4ND (Nguyen et al., 2022) separately computes dynamics along each axis of spatiotemporal data, while ConvS5 (Smith et al., 2023) restricts the recurrence to 1×1 convolutions. These choices reflect a structural barrier: composing spatial convolutions during a parallel scan causes kernel sizes to grow with each step, making the algorithm quadratic. We resolve this tension by lifting the convolutional SSM into the Fourier domain, where convolution composition becomes pointwise multiplication, restoring linear complexity while preserving full spatial coupling.

We call the resulting model the spectral convolutional SSM (SCSSM). We believe it is the first SSM that implements recurrent horizontal connections within a scalable and parallelizable architecture. On PathFinder and PathTracker, two tasks which require long-range spatial integration over static and dynamic stimuli, a one-layer SCSSM approaches human-level accuracy with fewer parameters and similar training speeds to Transformers (Figure 1). Moreover, we find early positive evidence for scaling SCSSMs: swapping the Transformers in a state-of-the-art edge vision architecture (the ShViT) with SCSSMs leads to better performance in a model with around 5% fewer parameters (74.98% vs. 73.66% accuracy).

The Spectral Convolutional SSM An SSM describes the continuous-time evolution of a hidden state $\boldsymbol{h}(t)$ (Equation (1)), which can be discretized as a recurrence (Equation (2)):

$$\boldsymbol{h}'(t) = \boldsymbol{A}\boldsymbol{h}(t) + \boldsymbol{B}\boldsymbol{x}(t) \quad (1a) \qquad \boldsymbol{h}_t = \overline{\boldsymbol{A}}\boldsymbol{h}_{t-1} + \overline{\boldsymbol{B}}\boldsymbol{x}_t \quad (2a)$$

$$\boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{h}(t) + \boldsymbol{D}\boldsymbol{x}(t) \quad (1b) \qquad \boldsymbol{y}_t = \boldsymbol{C}\boldsymbol{h}_t + \boldsymbol{D}\boldsymbol{x}_t \quad (2b)$$

Convolutional SSM. To model horizontal connectivity, we replace the matrix products in Equation (1) with spatial convolutions:

$$\frac{\partial \boldsymbol{h}}{\partial t}(r,t) = \boldsymbol{K}_h * \boldsymbol{h}(r,t) + \boldsymbol{K}_x * \boldsymbol{x}(r,t) \quad (3a)$$

$$\boldsymbol{y}(r,t) = \boldsymbol{K}_y * \boldsymbol{h}(r,t) \quad (3b)$$

where r indexes spatial position and $*$ denotes convolution. This is not scalable: composing $\boldsymbol{K}_h$ during a parallel scan makes complexity quadratic (Smith et al., 2023).

Spectral formulation. In the Fourier domain, convolution becomes pointwise multiplication. We apply the 2D FFT to the spatial dimensions, yielding:

$$\boldsymbol{A} = \boldsymbol{\Lambda}, \qquad \boldsymbol{B} = \mathrm{FFT} \circ \boldsymbol{K}_x, \qquad \boldsymbol{C} = \boldsymbol{K}_y \circ \mathrm{iFFT} \quad (4)$$

In submission to the *9th Conference on Cognitive Computational Neuroscience* (CCN 2026).

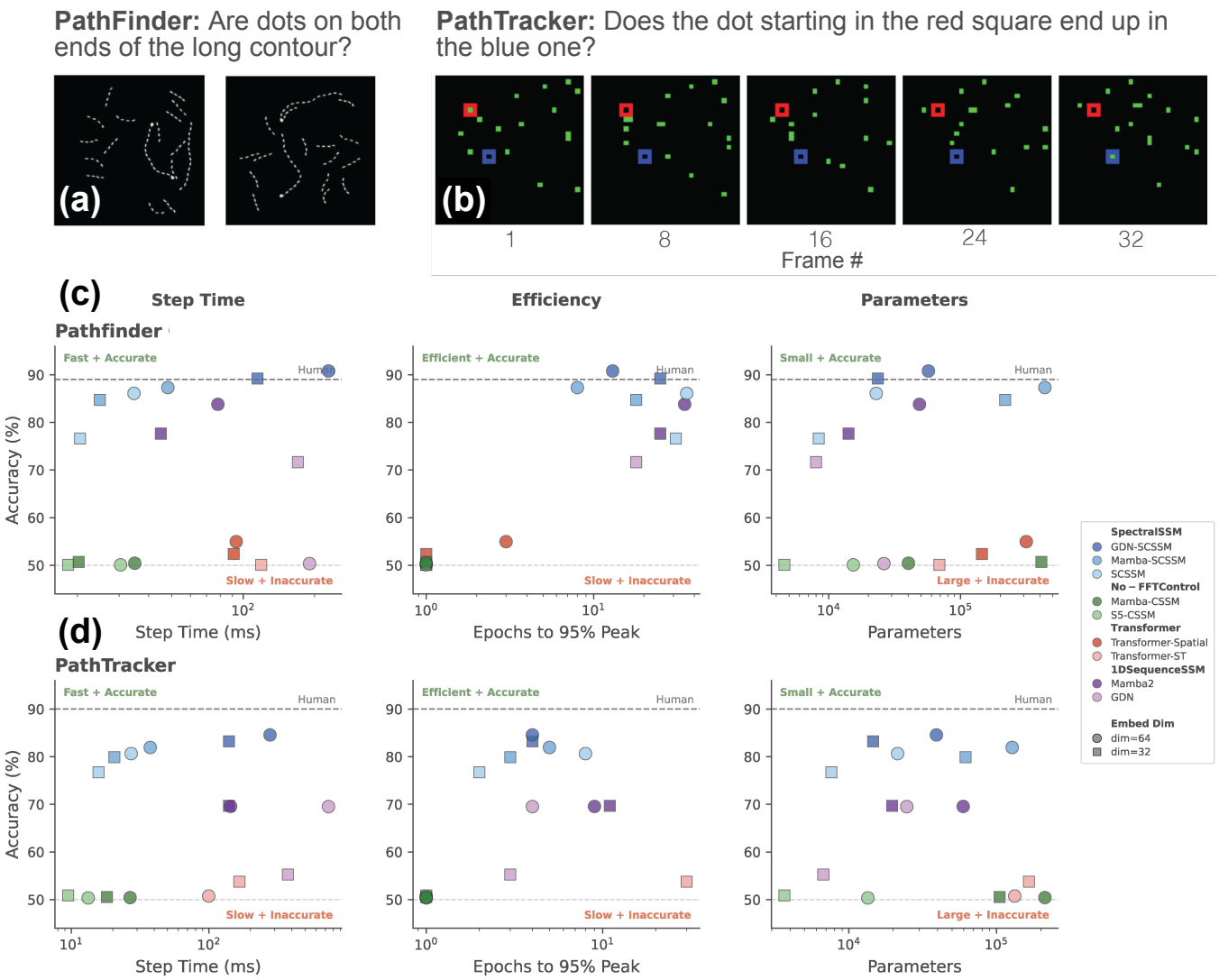

Figure 1: **A single layer of the SCSSM rivals humans on visual reasoning tasks.** We compare the SCSSM, leading SSMs, Transformers, and humans on two visual reasoning challenges: **(a)** PathFinder, and **(b)** PathTracker. **(c–d)** SCSSMs and its gated variants are more accurate and efficient than leading sequence models (Transformers and SSMs) and other recurrent convolution vision models. Removing the spectral formulation (No-FFTControl) reduces performance to chance, confirming that the spectral formulation is essential.

where $\mathbf{\Lambda}$ is a learnable diagonal in frequency space. The parallel scan thus reduces to scalar operations at each frequency, restoring linear complexity with full spatio-temporal coupling. On top of this core SCSSM, we also build versions with Mamba-style (Mamba-SCSSM) and Gated Delta Network-style gating (GDN-SCSSM).

Experiments We evaluate SCSSMs on **PathFinder** (Linsley et al., 2018), a static contour-tracing task (Figure 1a), and **PathTracker**, a dynamic 32-frame dot-tracking task (Figure 1b). We compare against Transformers (with spatial or spatiotemporal attention), sequence-based SSMs (Mamba2, GDN), and convolutional SSMs without the spectral formulation (Mamba-CSSM, S5-CSSM).

Results. SCSSM variants approach human accuracy on both tasks (Figure 1c–d). In contrast, non-spectral convolutional SSMs were at chance ($\sim$50%), confirming that the kernel growth problem prevents learning entirely. Transformers also struggle, while sequence-based SSMs are better, but still lag behind the SCSSM.

References

Bosking, W. H., Zhang, Y., Schofield, B., & Fitzpatrick, D. (1997). Orientation selectivity and the arrangement of horizontal connections in tree shrew striate cortex. *Journal of Neuroscience*, *17*(6), 2112–2127.

Field, D. J., Hayes, A., & Hess, R. F. (1993). Contour integration by the human visual system: Evidence for a local "association field". *Vision Research*, *33*(2), 173–193.

Gilbert, C. D., & Li, W. (2013). Top-down influences on visual processing. *Nature Reviews Neuroscience*, *14*(5), 350–363.

Gilbert, C. D., & Wiesel, T. N. (1989). Columnar specificity of intrinsic horizontal and corticocortical connections in cat visual cortex. *Journal of Neuroscience*, *9*(7), 2432–2442.

Gu, A., Goel, K., & Ré, C. (2022). Efficiently modeling long sequences with structured state spaces. *International Conference on Learning Representations*.

Linsley, D., Ashok, A., Govindarajan, L., Liu, R., & Serre, T. (2020). Stable and expressive recurrent vision models. *Advances in Neural Information Processing Systems*, *33*, 10456–10467.

Linsley, D., Feng, P., & Serre, T. (2025). Better artificial intelligence does not mean better models of biology. *Trends Cogn. Sci.*

Linsley, D., Kim, J., Veerabadran, V., Windolf, C., & Serre, T. (2018). Learning long-range spatial dependencies with horizontal gated recurrent units. *Advances in Neural Information Processing Systems*, *31*.

Mély, D. A., Linsley, D., & Serre, T. (2018). Towards a general model of surround modulation in the visual cortex. *Computational Brain & Behavior*, *1*, 16–32.

Nguyen, E., Goel, K., Gu, A., Downs, G., Shah, P., Dao, T., Baccus, S., & Ré, C. (2022). S4ND: Modeling images and videos as multidimensional signals with state spaces. *Advances in Neural Information Processing Systems*, *35*.

Smith, J. T., De Mello, S., Kautz, J., Linderman, S., & Byeon, W. (2023). Convolutional state space models for long-range spatiotemporal modeling. *Advances in Neural Information Processing Systems*, *36*.

Tay, Y., Dehghani, M., Abnar, S., Shen, Y., Bahri, D., Pham, P., Rao, J., Yang, L., Ruder, S., & Metzler, D. (2021). Long range arena: A benchmark for efficient transformers. *International Conference on Learning Representations*.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*.